

David Dangerfield

Principal Frontend Engineer — Regulated and Mission-Critical Systems

Payment Infrastructure | Healthcare | Medical Devices | 20+ Years

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SUMMARY

Principal frontend engineer with 20+ years of software engineering experience, specializing in mission-critical systems where correctness, reliability, and accountability are non-negotiable. Career includes healthcare platforms, medical devices, FDA clinical trial software, patient monitoring systems, and payment infrastructure handling billions in annual revenue. Deep experience working under HIPAA compliance, FDA regulatory protocols, SOPs, and audit trail requirements. Strong computer science foundation (MS, Purdue) spanning frontend, backend, test architecture, and systems engineering. Current with AI-assisted development tooling, applying it to accelerate delivery while maintaining the standards that regulated environments demand. My allegiance is to the business: I recognize and reject proposals that serve developer interests but bring little business value.

SKILLS

Frontend: TypeScript, JavaScript, Vue.js, React, Angular, HTML, CSS, Tailwind, Bootstrap, D3.js

Backend: Node.js, GraphQL, REST APIs, Java, C#, C++, RDBMS, Message Queues

Testing: Vitest, Playwright, Cypress, Jest, xUnit

Tools: Git, Vite, Storybook, Webpack, GitHub, Azure DevOps

EXPERIENCE

Principal Frontend Engineer

March 2025 - Present

[Patterson Companies, Inc.](#)

- Rewrote the bill pay system for dental and veterinary sales sites serving practices across North America, supporting a multi-billion dollar annual revenue stream
- Implemented pixel-perfect TypeScript/Vue.js frontend against Figma designs with rigorous attention to correctness in high-consequence financial workflows
- Extended and improved Patterson's proprietary UI component library, improving consistency and maintainability across properties

Tech: TypeScript, Vue.js, Node.js, Vite, Vitest, Storybook, Git, Azure DevOps

Senior Frontend Engineer (part-time)

February 2018 - March 2024

[Nurocor - Digitalized Clinical Development](#)

- Built a structured authoring platform for FDA clinical trial documentation, enabling compliant document templates, formal review workflows, and complete audit trails on every edit
- Implemented audit trail functionality ensuring full traceability, a core requirement for FDA regulatory submission environments
- Platform recognized by the TransCelerate BioPharma ClinEco Protocol Review Challenge as Best Integrated Protocol Authoring and Review Solution

Tech: Vue.js, Vuetify, Storybook, GraphQL, Git

Lead Frontend Developer

July 2022 - March 2025

[Life Time Inc.](#)

- Built web and app experiences for one of the largest consumer fitness platforms in North America, driving hundreds of thousands of new accounts and app installs
- Modernized legacy frontend dependencies, reducing technical debt across high-traffic consumer properties

Tech: TypeScript, Vue.js, Bootstrap, Vite, Vitest, Playwright, GitHub

Principal Frontend Engineer Consultant

January 2021 - July 2022

[NextEra Analytics](#)

- Built decision-support tools helping energy sector leaders direct billions of dollars in annual capital spending, replacing manual spreadsheet workflows with scalable web applications
- Developed executive dashboards and data visualizations translating complex capital allocation data into actionable business intelligence

Tech: Vue.js, GraphQL, eCharts, Storybook, Git

Senior Frontend Consultant (part-time)

September 2020 - February 2022

[Intricately \(Acq. by HG Insights\)](#)

- Engaged directly by CTO to elevate frontend engineering practices across an internationally distributed development team; proposed and sold a part-time model delivering principal-level leadership at reduced cost
- Contributed to a B2B cloud spend intelligence platform subsequently acquired by HG Insights

Tech: TypeScript, Vue.js, Git

Senior Frontend Engineer

September 2018 - January 2021

SmugMug

- Developed and maintained SmugMug.com, one of the internet's earliest subscription services still in operation, hosting billions of photos for millions of customers worldwide
- Replaced legacy custom payment processing with Stripe integration, modernizing financial infrastructure for a high-volume consumer platform

Tech: JavaScript, React, Tailwind, GraphQL, Git, continuous deployment

Frontend Developer, Watson Health Consulting Services

November 2015 - August 2018

IBM Services

- Developed an early AI-powered pharmaceutical research tool built on IBM Watson's natural language processing capabilities, ingesting scientific literature to help researchers understand relationships between biomedical entities
- Built clinical decision support applications for healthcare workers including diabetes treatment recommendations and patient history visualization, under HIPAA compliance and regulatory protocols
- Delivered cognitive AI applications for financial clients including a call center efficiency platform for a top 50 US bank
- Co-founded FED@IBM Rochester, an internal frontend developer community of 30+ engineers hosting twice-monthly sessions on HTML, CSS, JavaScript, accessibility, UI automation testing, and developer tooling

Tech: Vue.js, Angular, JavaScript, HTML, CSS

Senior Web Application Developer

August 2013 - October 2015

Preventice Solutions (now Boston Scientific)

- Developed frontend and backend systems supporting BodyGuardian, an FDA-cleared wearable cardiac monitoring device transmitting real-time patient data to clinical care teams
- Built the Notification Service delivering cardiac monitoring alerts to clinical staff, a critical component in a live patient care pipeline requiring high reliability
- Worked under formal FDA/HIPAA compliance protocols including SOPs, audit trails, required training, and documented sign-offs throughout the development lifecycle

Tech: JavaScript, SAML SSO, ASP.NET, C#, ActiveMQ, Java, Angular

Software Engineer / Team Lead / Test Architect

May 2002 - August 2013

IBM Systems

- Ported IBM's Directory Server (LDAP) to the IBM i platform (iSeries/AS400) in C++, contributing to a commercially shipped product on one of the most reliable enterprise platforms in the industry
- Architected test infrastructure and led functional testing across 12-15 person teams spanning Rochester and Beijing, coordinating regression automation across multiple simultaneous releases
- Automated software update packaging in a scripting initiative that eliminated 500+ person-hours of manual effort annually and enabled self-service builds across multiple teams

Tech: C++, Java, JavaScript, HTML, CSS, Perl, IBM Db2

EDUCATION

Master of Science, Computer Science | Purdue University, 2002

Cryptography, Distributed Databases, Networking, Operating Systems

Bachelor of Science, Computer Science | Brigham Young University, 2000

Networking, Artificial Intelligence and Neural Networks, Computer Graphics